

**Evaluating the spatio-temporal dynamics of low- and mid-trophic
level resources in the Azores:
implications for the new MPA network and cetacean populations**

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Introduction

Marine food webs:

Low trophic levels: NPP and zooplankton

Mid-trophic levels: micronekton

Apex predators: cetaceans

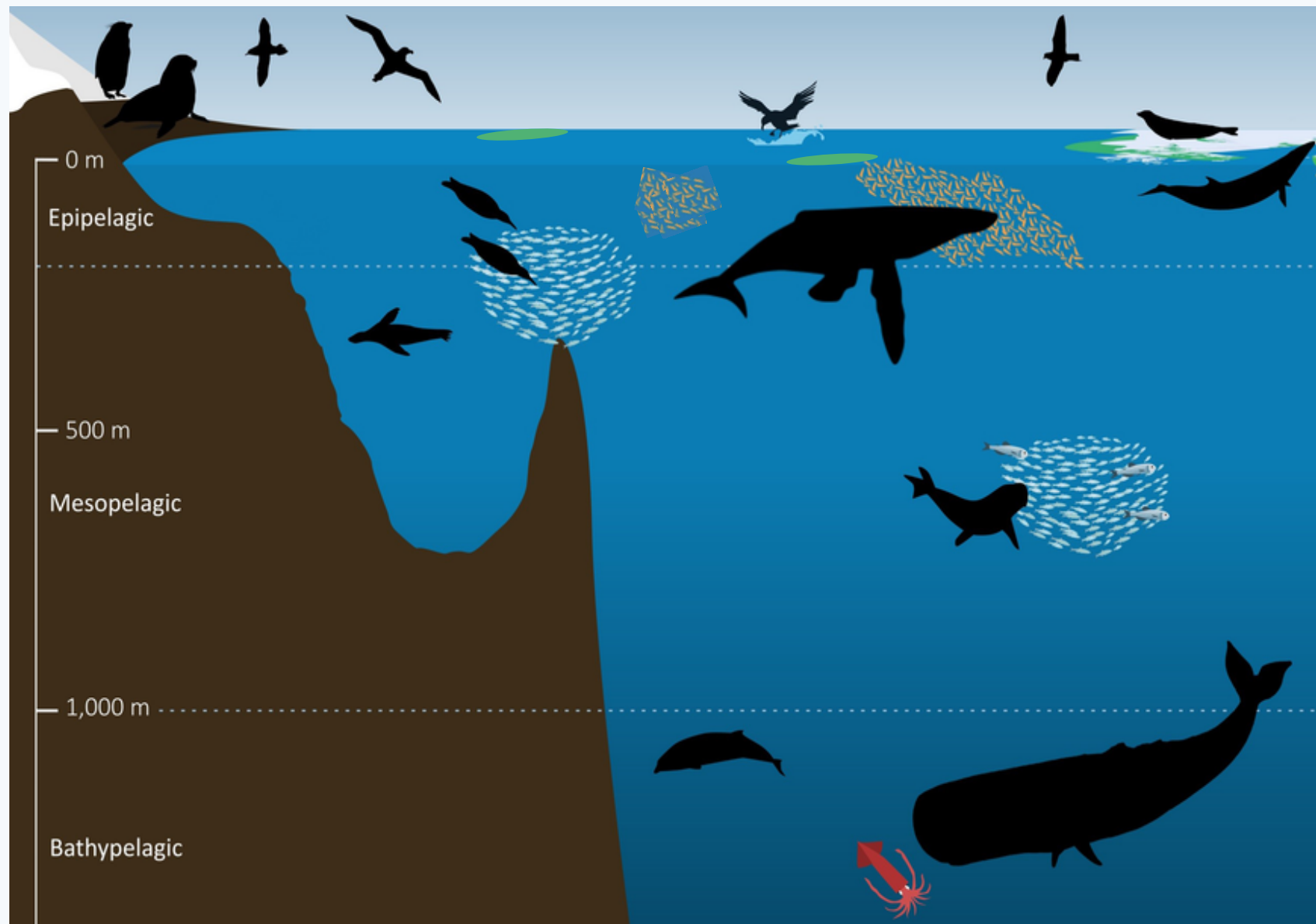


Figure 1 : Marine trophic relationship

Sampling methods

NPP:

Satellite remote sensing
Satellite-derived NPP models

Zooplankton:

Net sampling
Acoustic and modeling

Apex predators:

Sightings and tracking data

Micronekton:

Sound scattering

—> **Ecological model**

Introduction

SEAPOODYM ecological model:

(Lehodey & Senina, 2009; Lehodey et al., 2010)

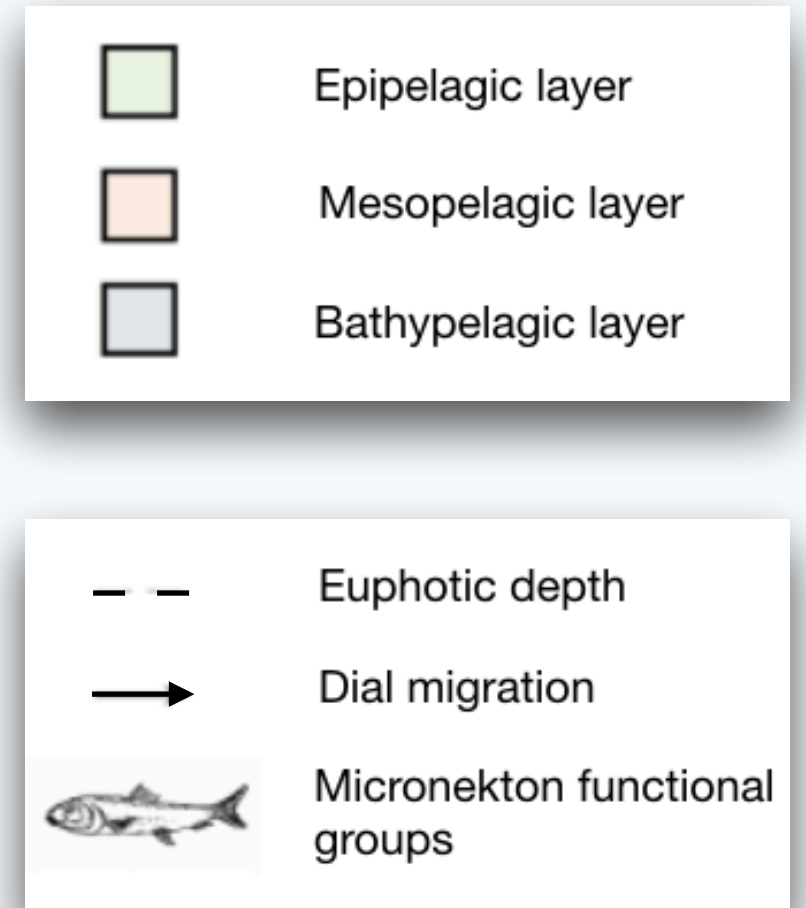
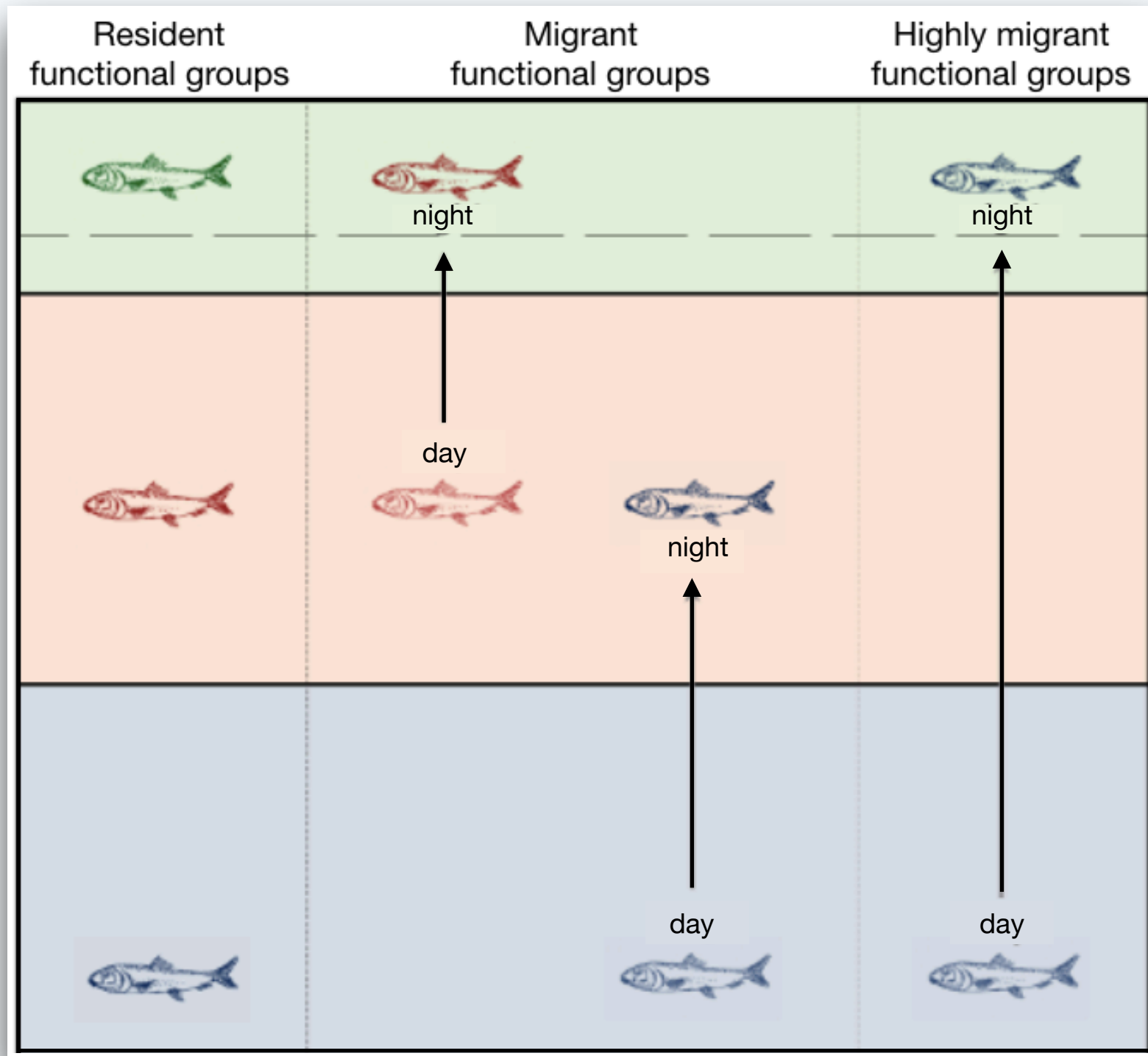


Figure 2 : SEAPOODYM six functional groups of micronekton. Adapted from Lambert et al., (2014).

Objectives:

Investigate the **spatio-temporal distribution** of low- and mid-trophic level **biomass** in the Azores

- Biomass and productivity of the **MPA network**
- Influence of productivity on **cetacean populations**

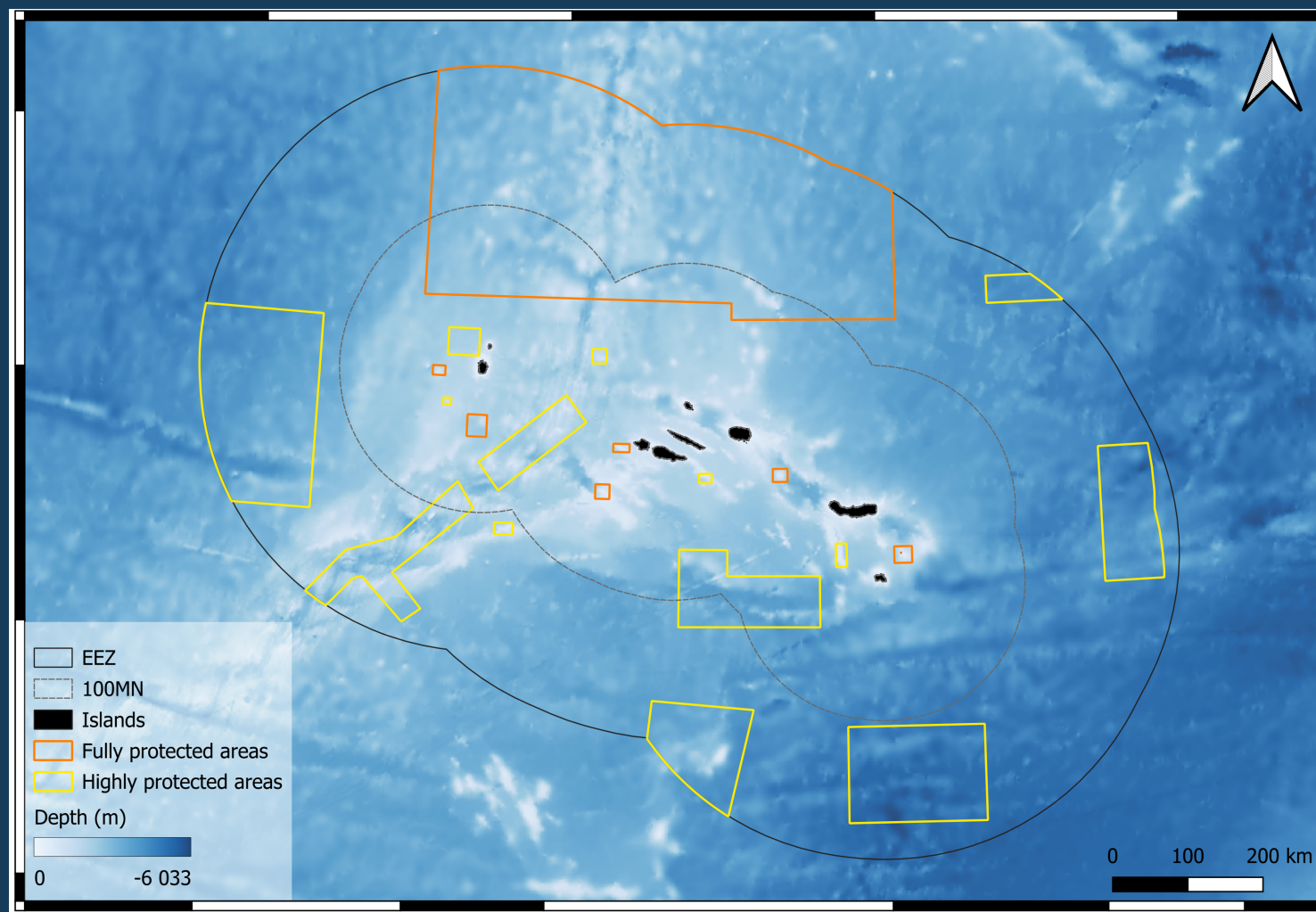


Figure 3: Map of the Azores archipelago and the MPA network

Materials and Methods

1. Epipelagic resources

- SEAPODYM-LMTL data (1998-2023)
- NPP + zooplankton + micronekton

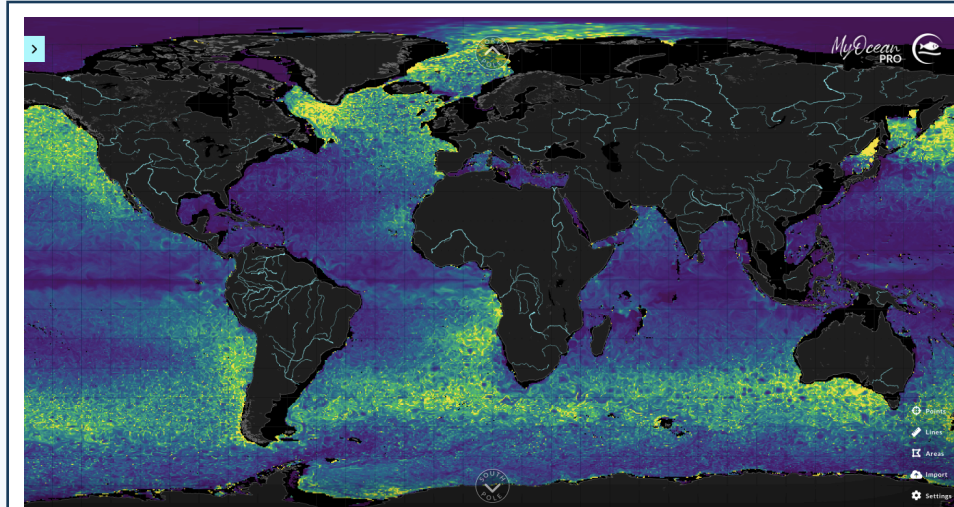


Figure 4: SEAPODYM spatial coverage

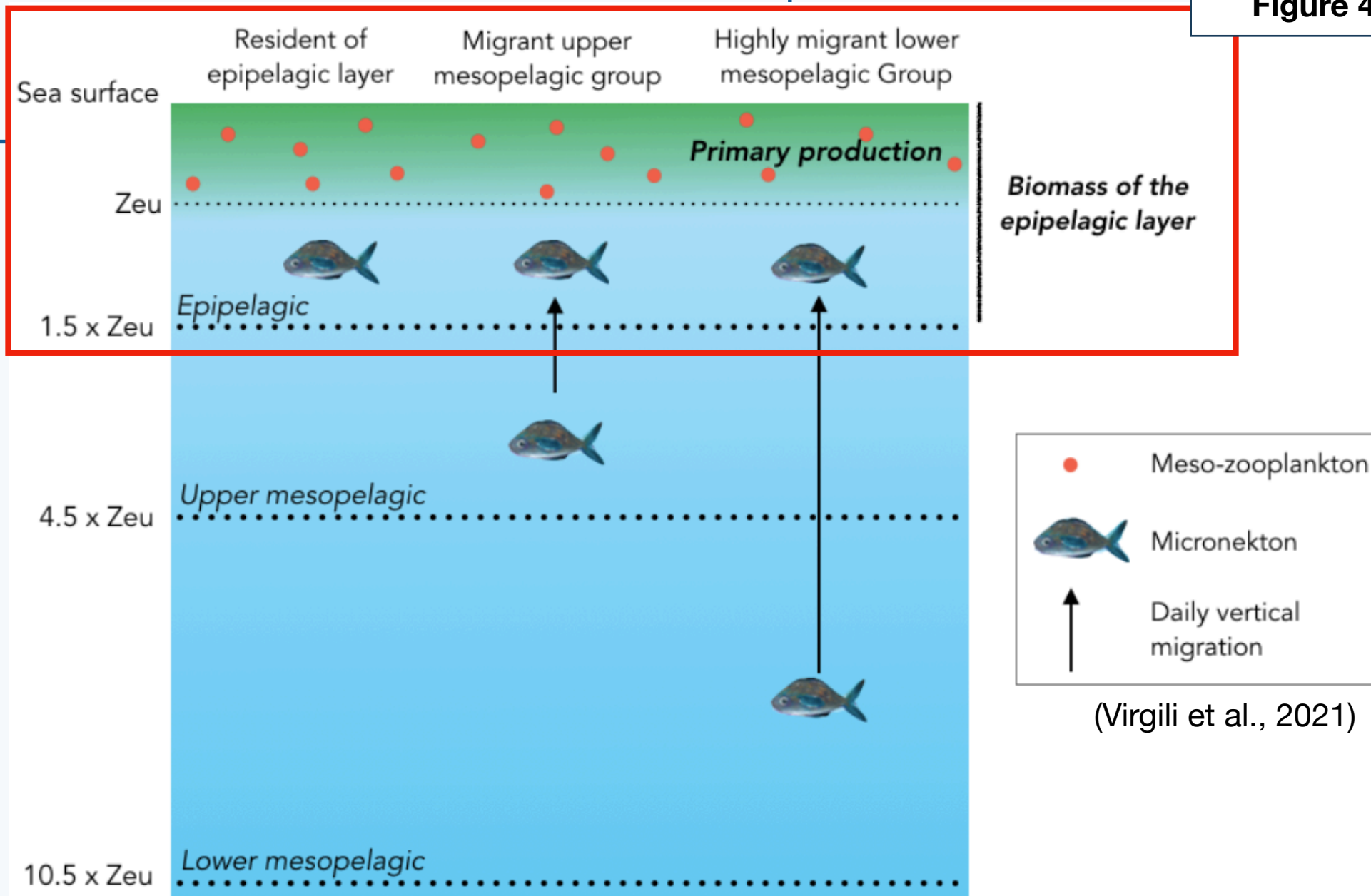
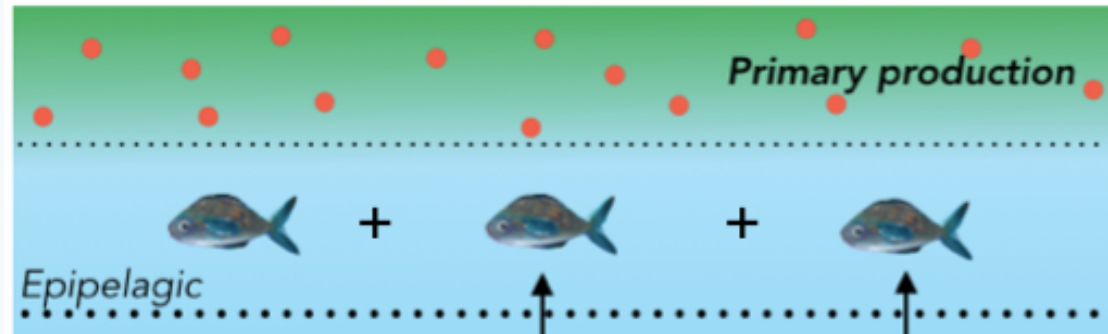


Figure 5: Biomass in the epipelagic layer during the night in SEAPODYM-LMTL model

Materials and Methods

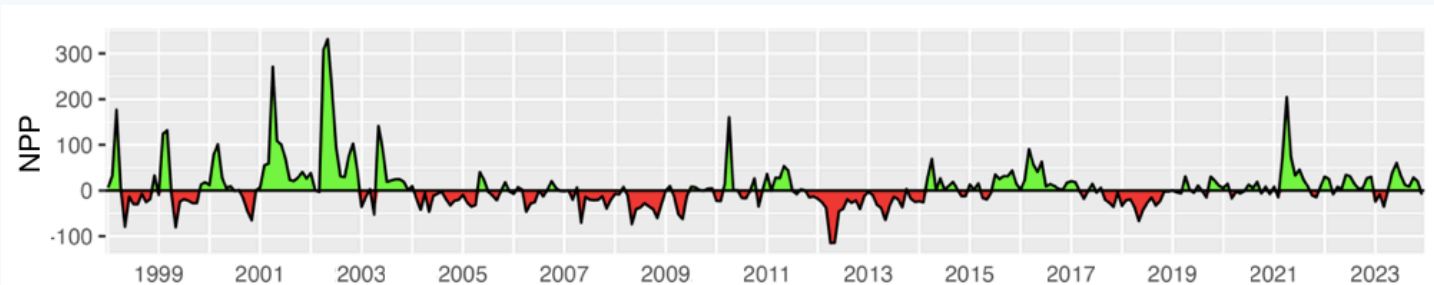
1. Epipelagic resources

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- NPP + zooplankton + micronekton



2. Temporal study (1998-2023)

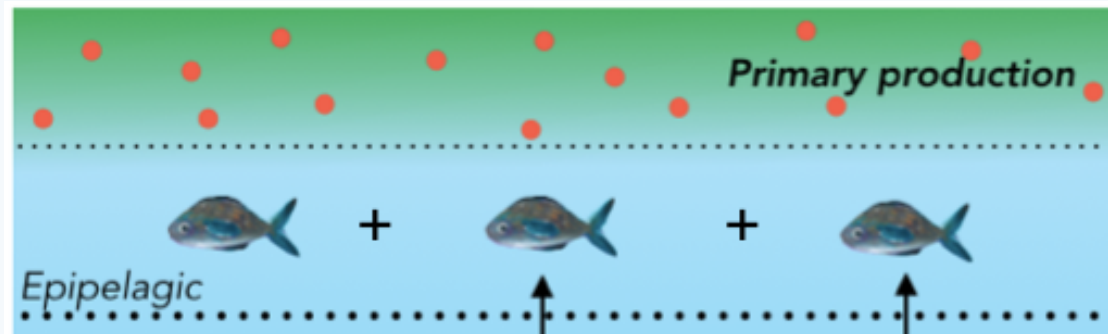
- Annual
- Seasonal variations + phenology



Materials and Methods

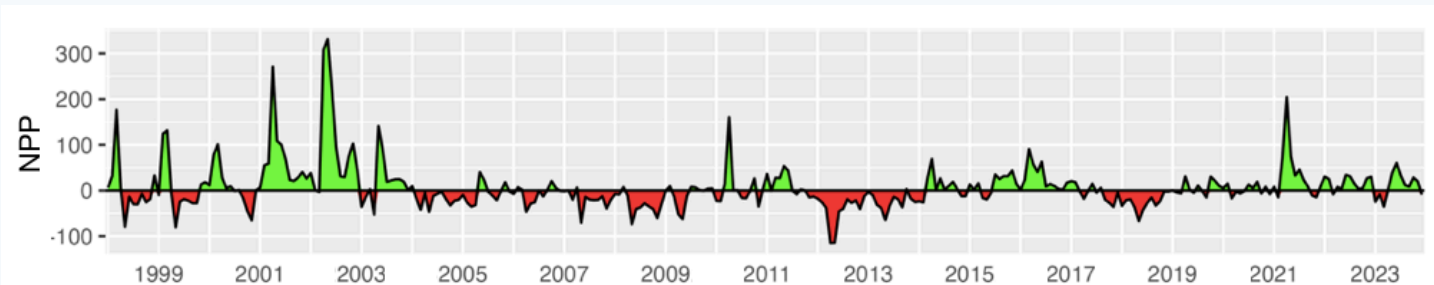
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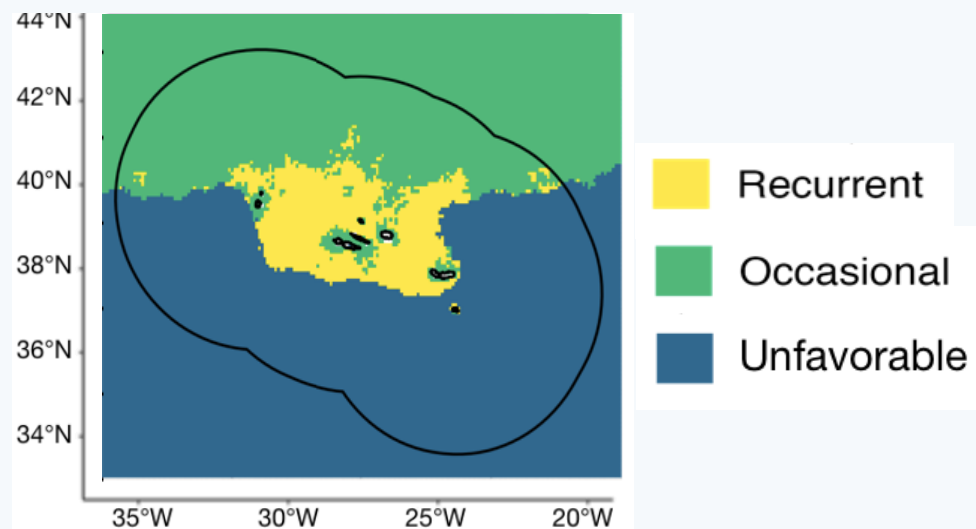
2. Temporal study (1998-2023)

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3. Spatial study

- Biomass distribution and productivity zones



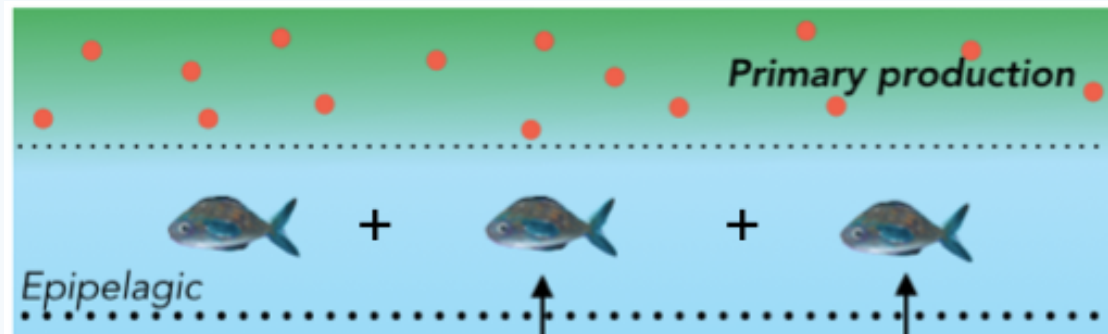
- median > overall median ; IQR < overall IQR
- median > overall median ; IQR > overall IQR
- median < overall median

- **Climatological study** : Biomass compared with an overall median for each month (1998-2023)
- **Contemporaneous study** : Biomass compared with a monthly median for each month of each year (1998-2023)

Materials and Methods

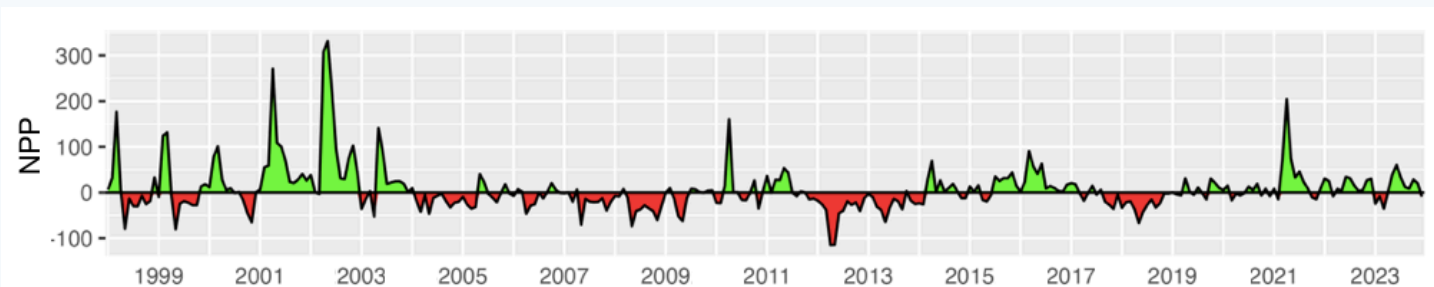
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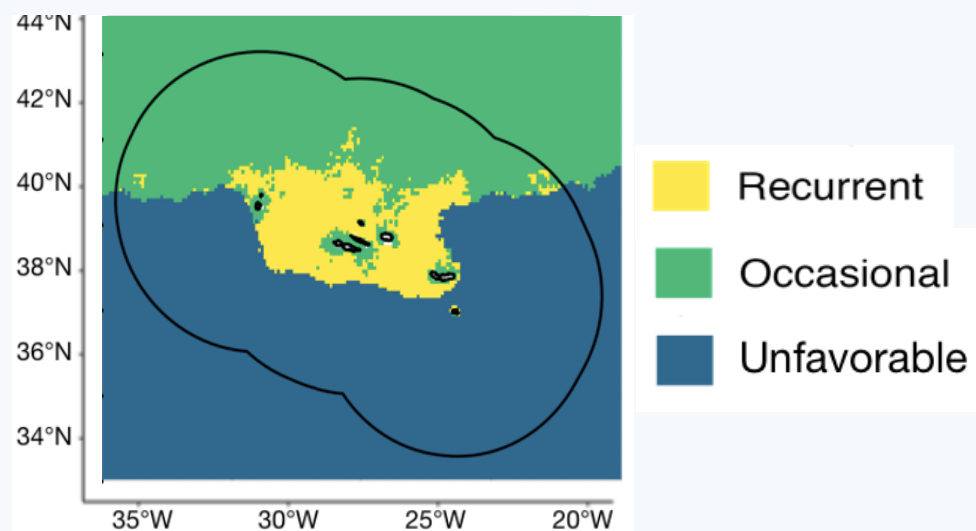
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- Annual
- Seasonal variations + phenology



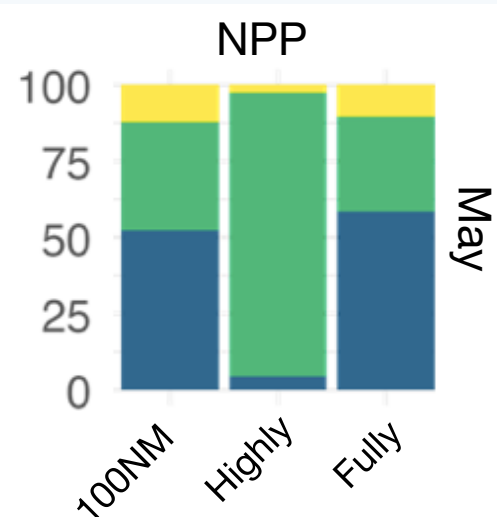
3. Spatial study

- Biomass distribution and productivity zones



4. MPA

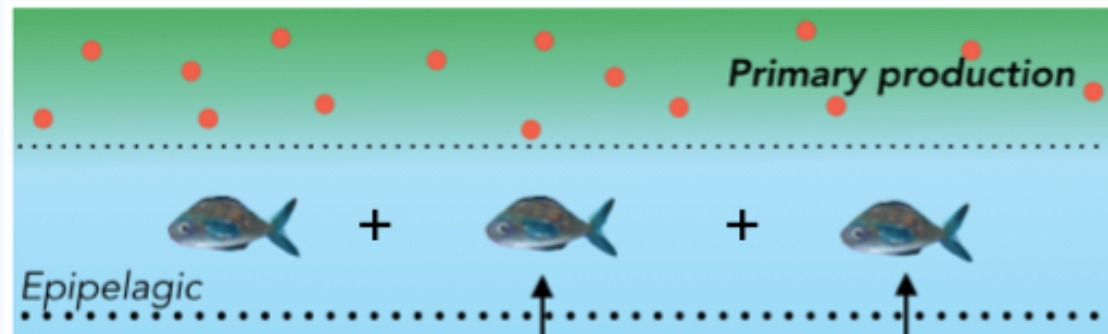
- 3 levels of protection
- Biomass + productivity



Materials and Methods

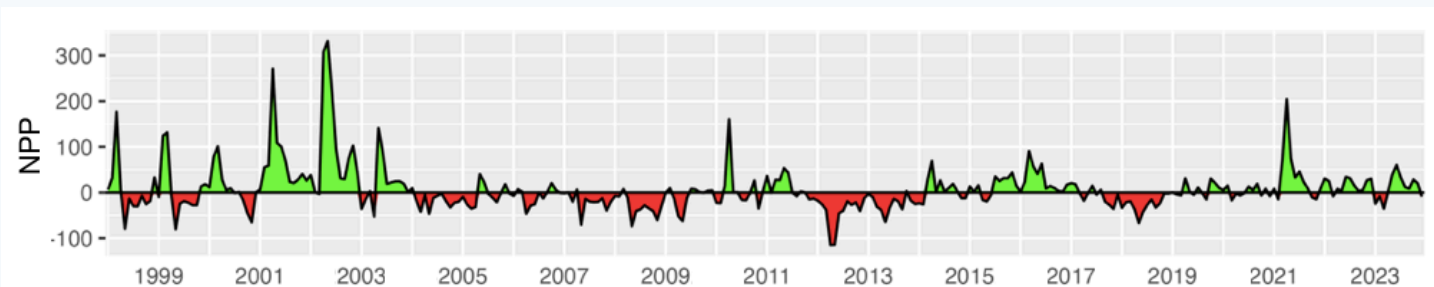
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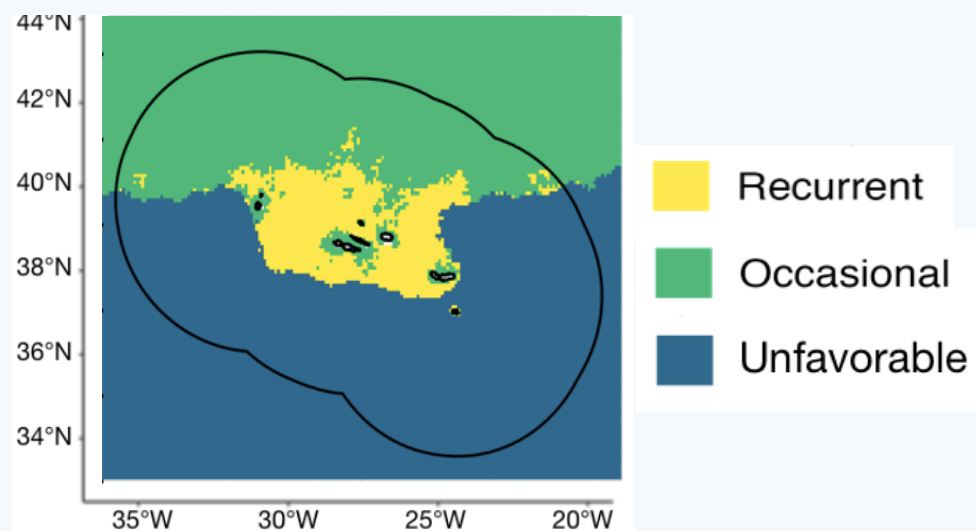
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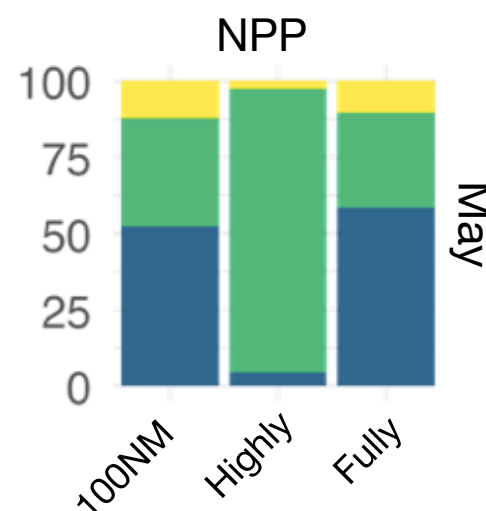
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- Biomass distribution and productivity zones



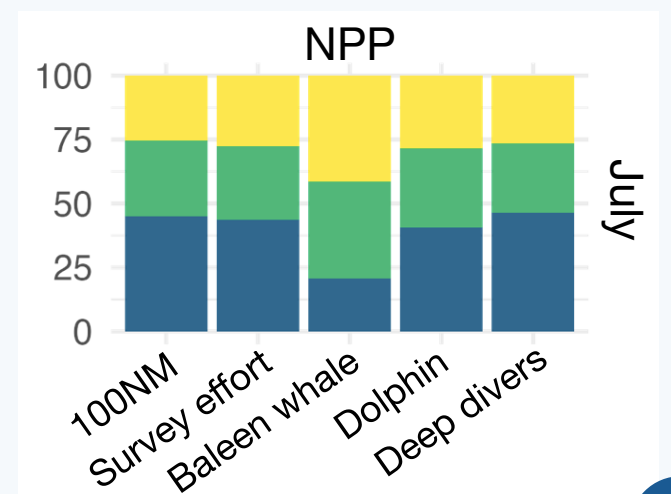
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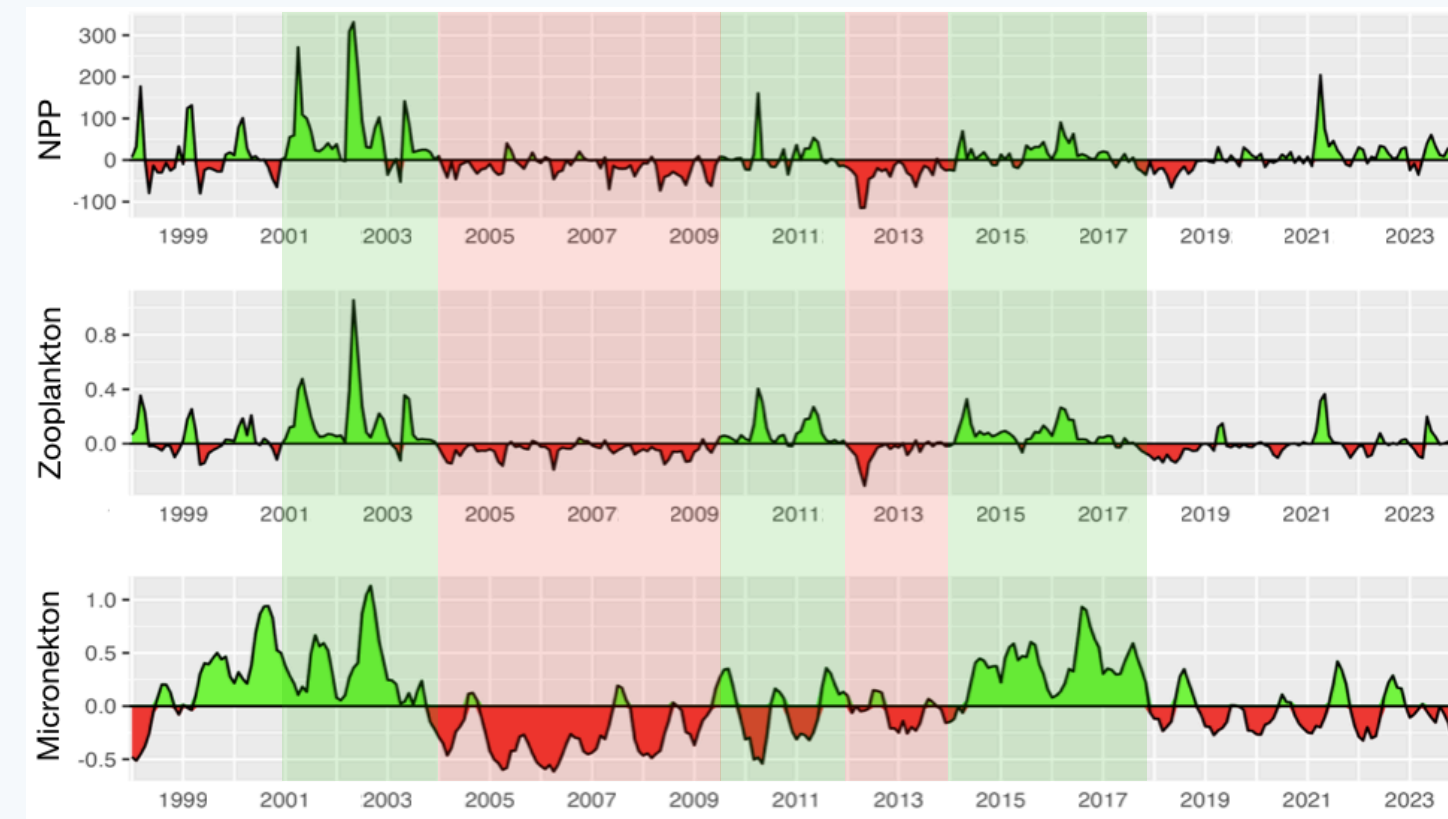
5. Cetaceans

- POPA data (2001-2015)
- 3 functional groups
- Productivity



Results

Temporal analysis



Annual

- No global trend concerning biomass and phenology
- Important inter-annual anomalies

Figure 6: Temporal variability in biomass anomalies for NPP, zooplankton and micronekton (1998-2023)

Seasonal

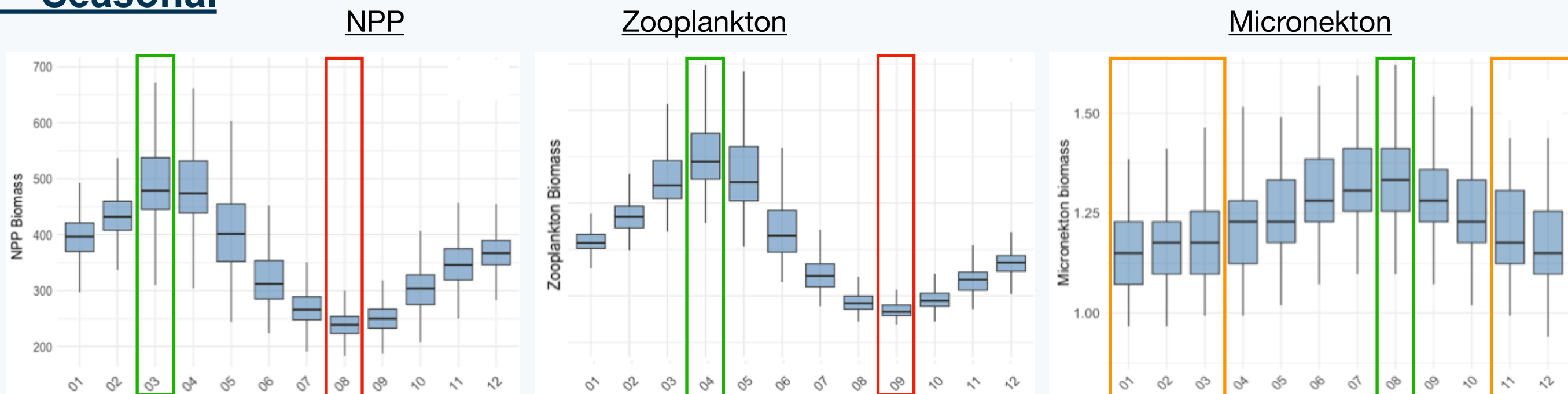


Figure 7 : Monthly biomass of NPP, zooplankton and micronekton (1998-2023)

Spatial analysis and productivity

- Recurrent: high and stable biomass
- Occasional: high and unstable biomass
- Unfavorable: low biomass

Productive waters
around the islands

High variable
productivity in the
Northern part

Low productivity
below 37°N

Results

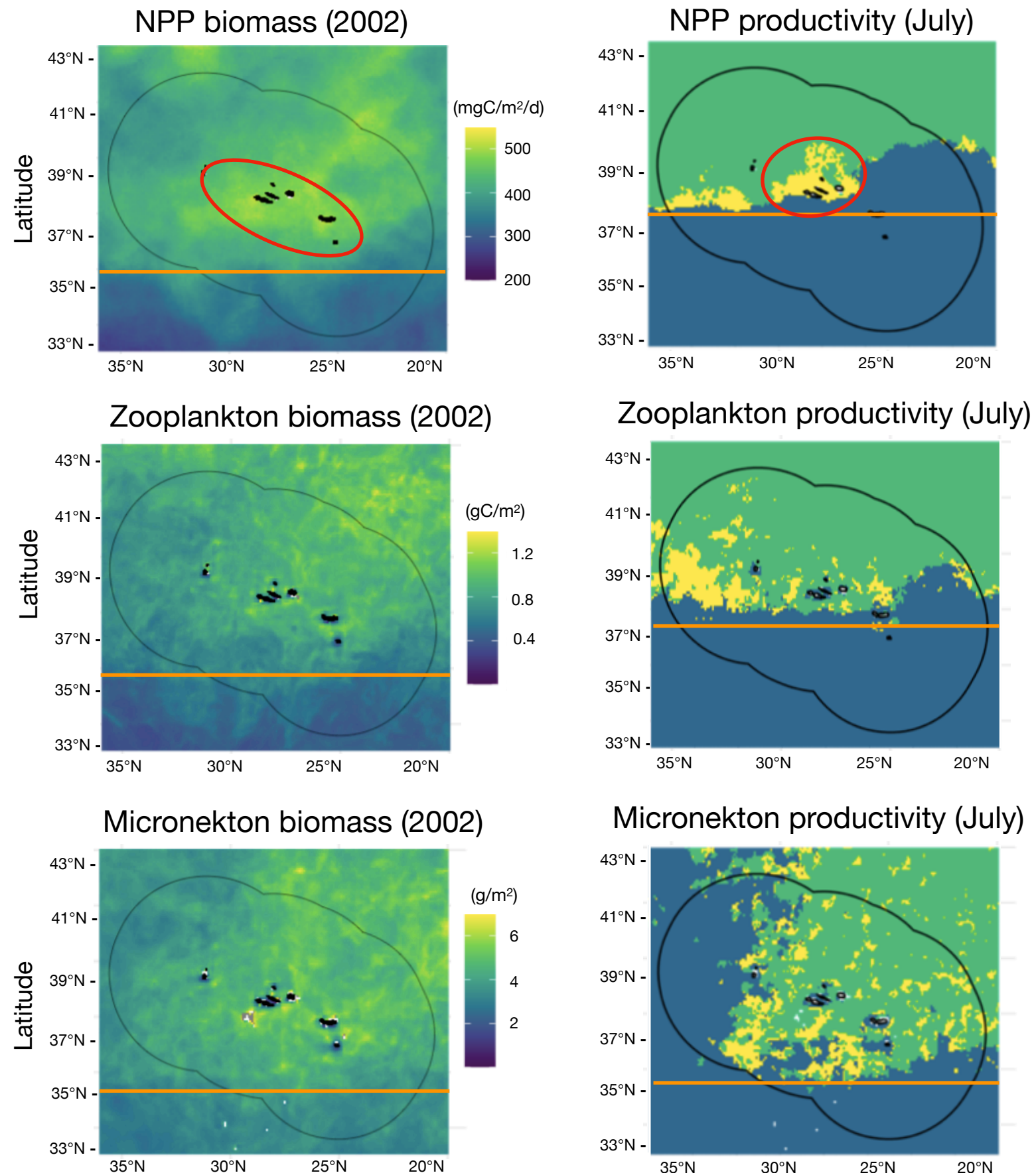


Figure 8: Spatial distribution of the biomass and productivity of the three variables

Results

MPA and productivity

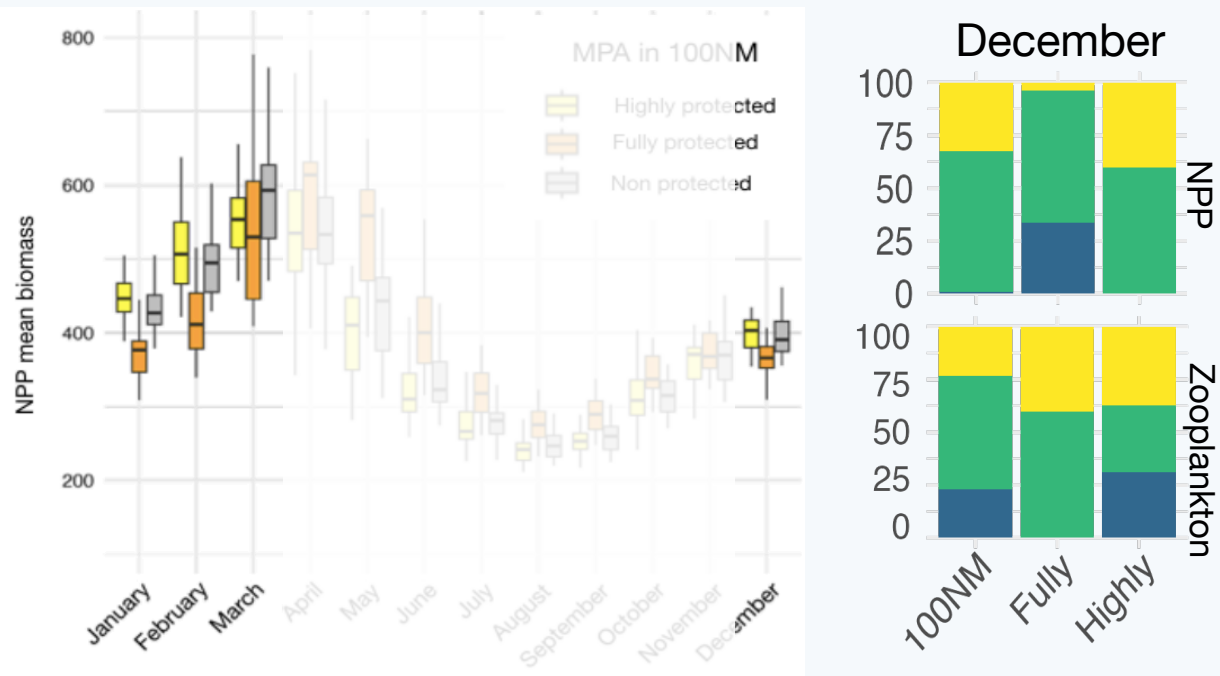


Figure 9 : Biomass and productivity of each type of MPA for NPP

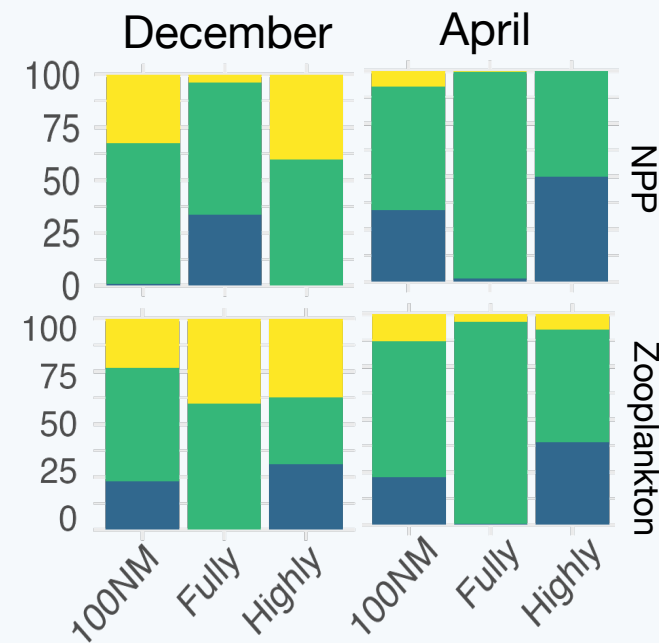
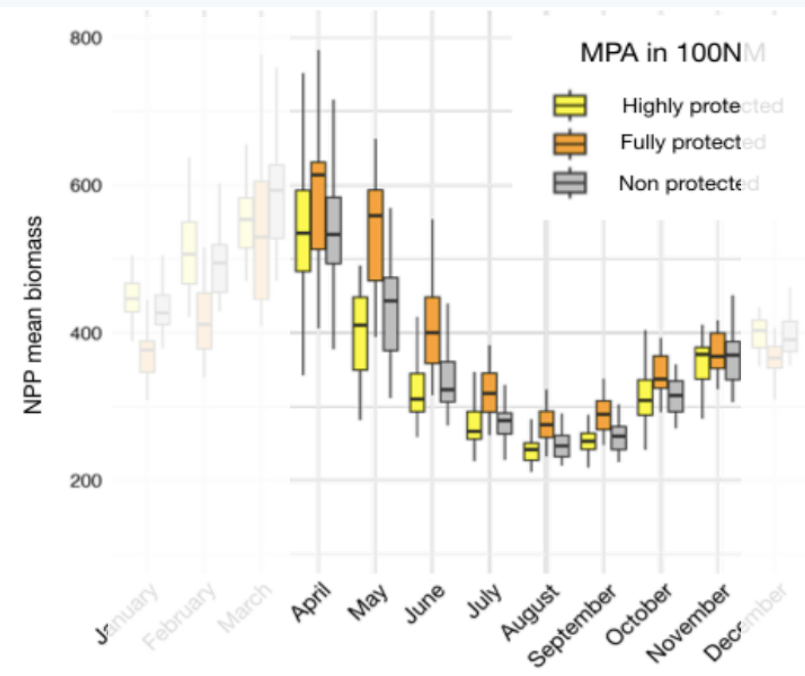
- **NPP and zooplankton:**

From December to March:
non and highly protected

Figure 10 : Biomass and productivity of each type of MPA for micronekton

Results

MPA and productivity



- **NPP and zooplankton:**

From December to March:
non and highly protected

From April to November:
fully protected

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Figure 10 : Biomass and productivity of each type of MPA for micronekton

Results

MPA and productivity

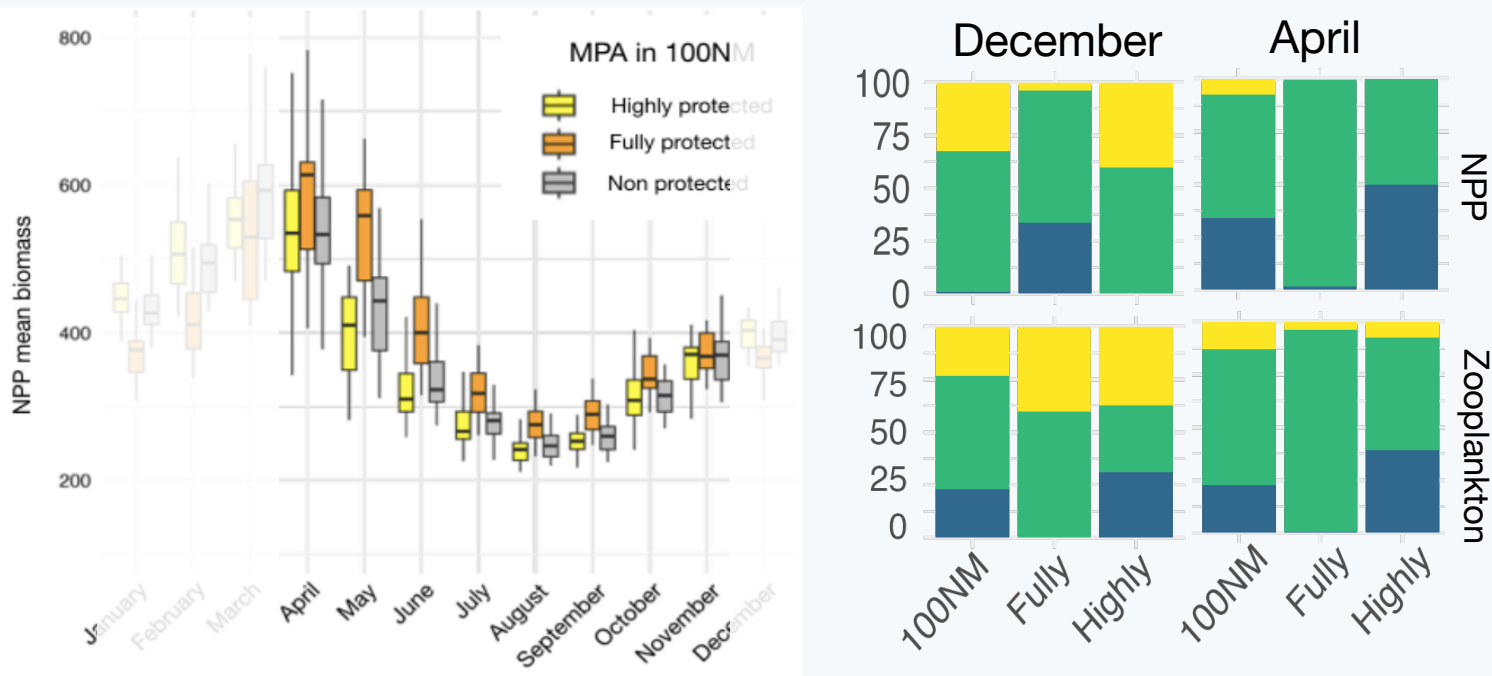


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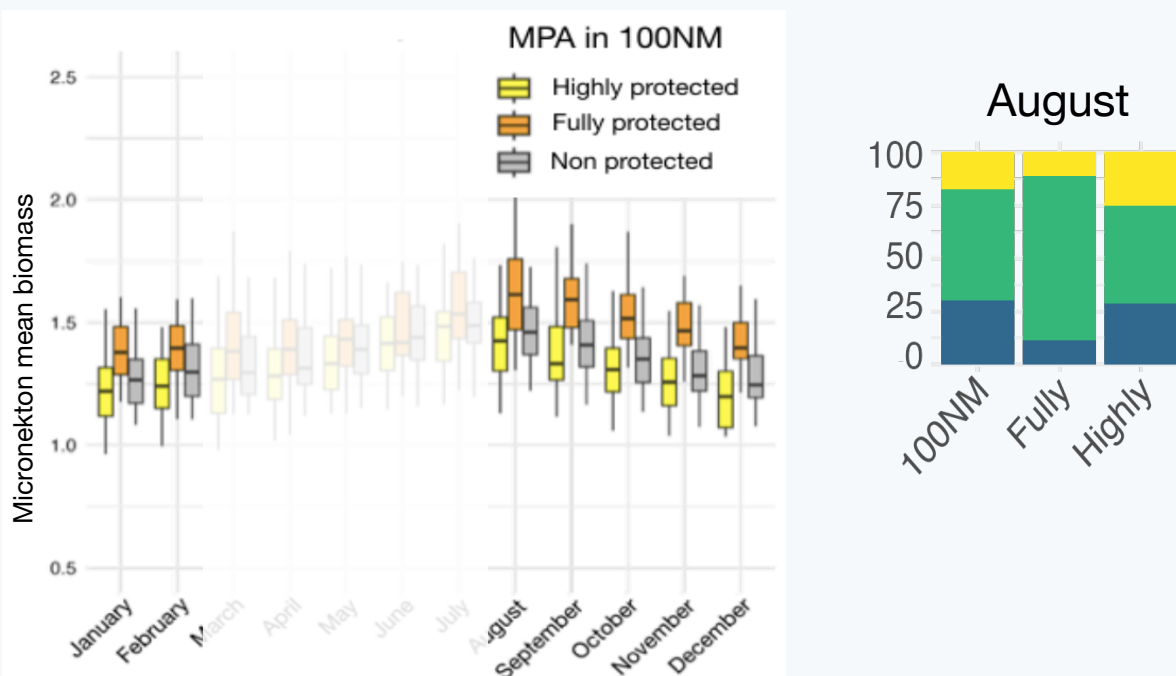


Figure 10 : Biomass and productivity of each type of MPA for micronekton

- **NPP and zooplankton:**

From December to March:
non and highly protected

From April to November:
fully protected

- **Micronekton:**

From August to February:
fully protected

Results

MPA and productivity

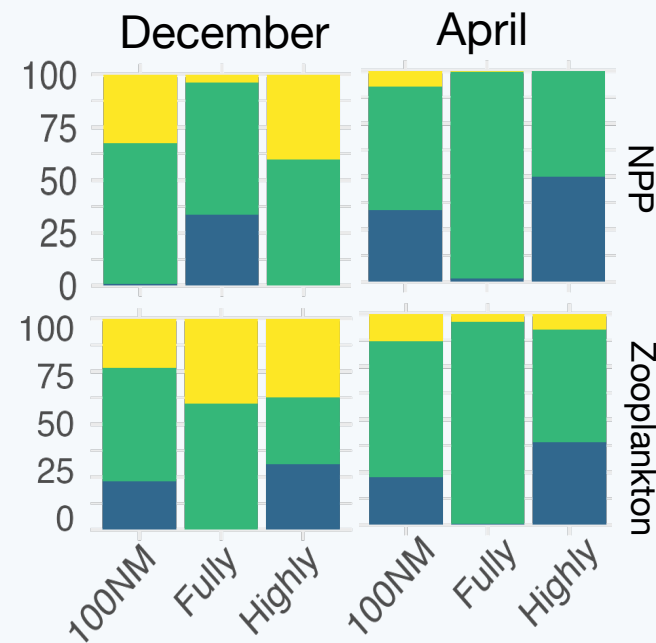
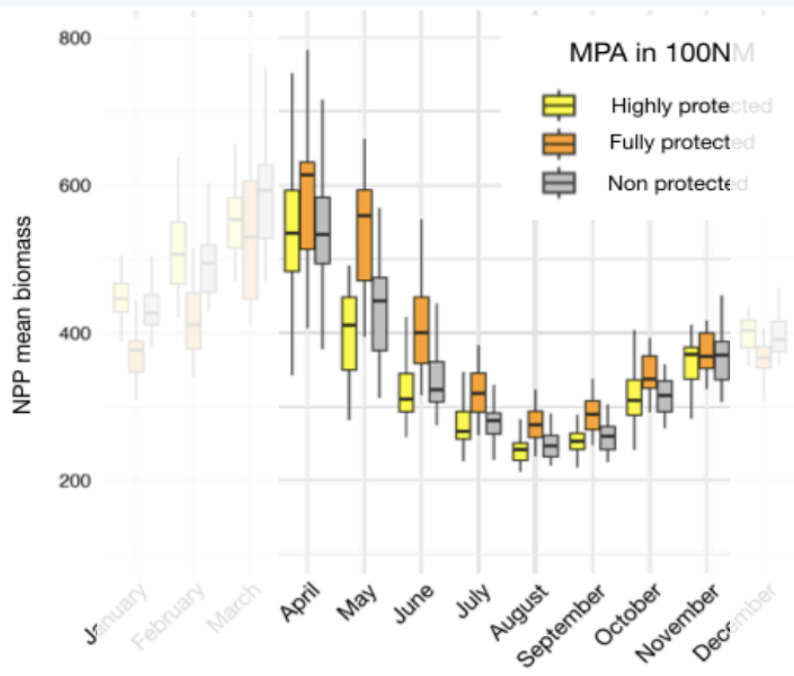


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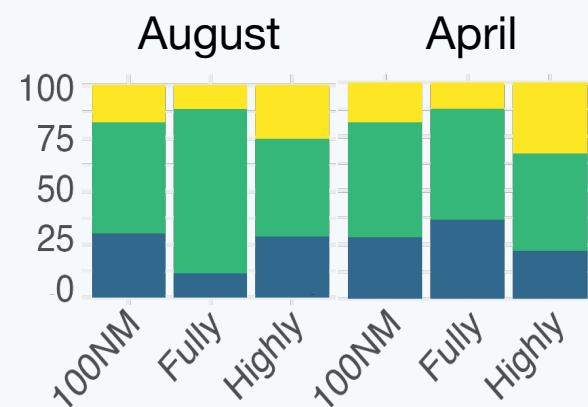
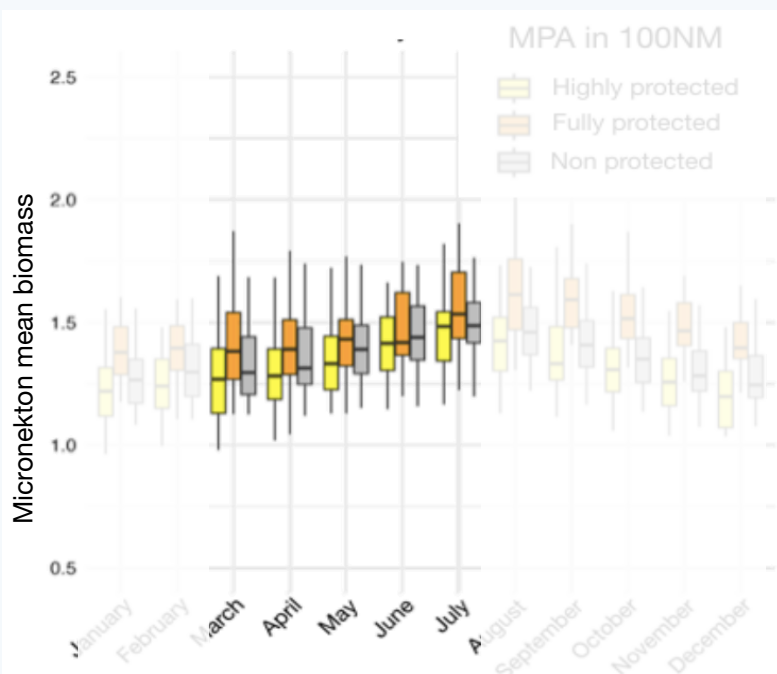


Figure 10 : Biomass and productivity of each type of MPA for micronekton

- **NPP and zooplankton:**

From December to March:
non and highly protected

From April to November:
fully protected

- **Micronekton:**

From August to February:
fully protected

From April to July:
non and highly protected

Results

Cetaceans and productivity

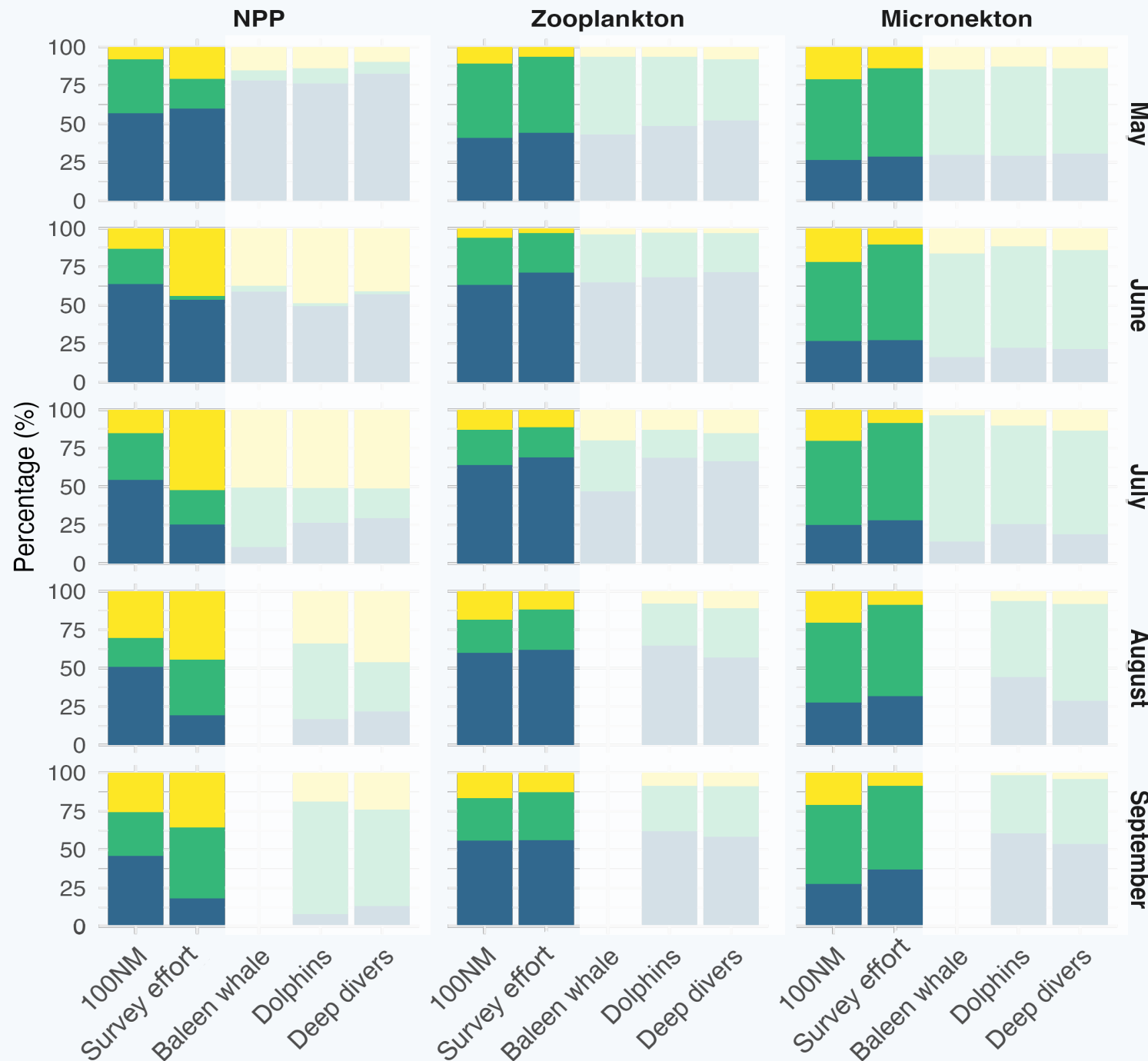


Figure 11: Proportion of each productivity zone for each variable within the 100 NM area, the vessel effort, and the three functional groups for the climatological study

Survey effort and 100NM:

NPP:

More productive zones for the survey effort

Zooplankton and micronekton:

Same proportions

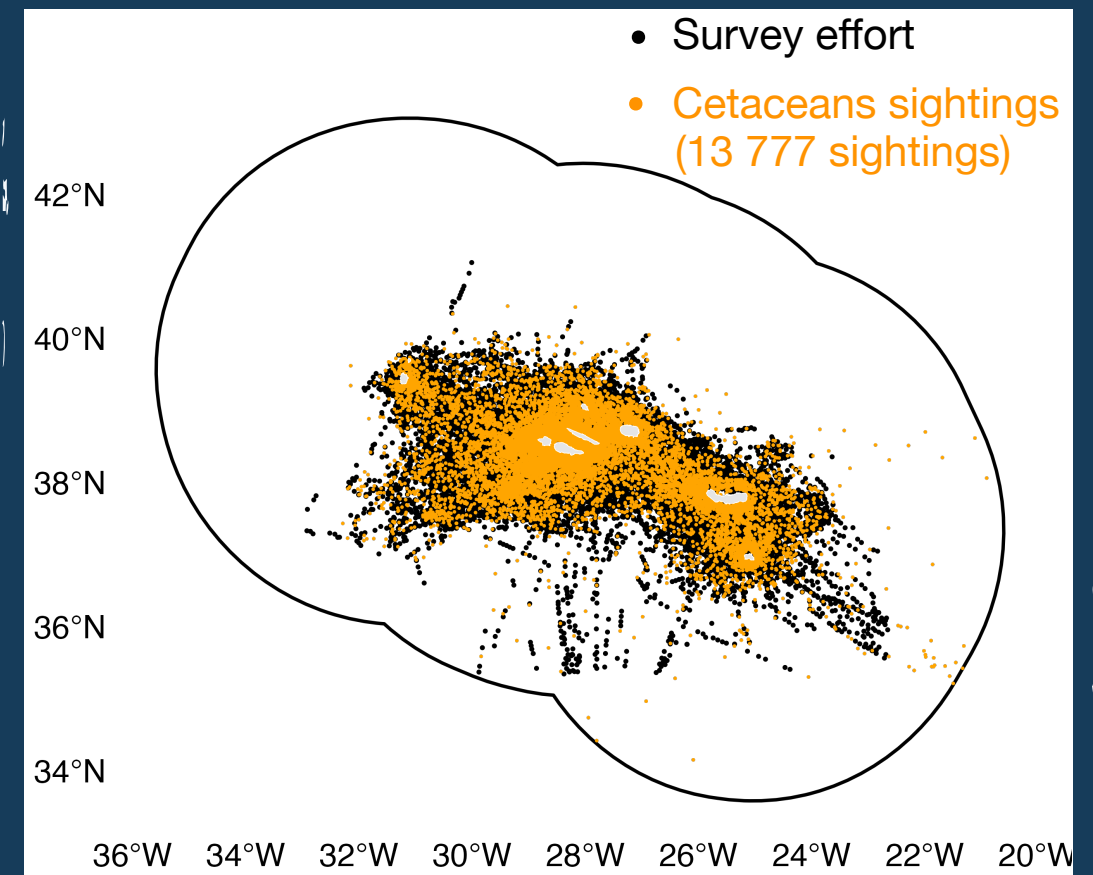
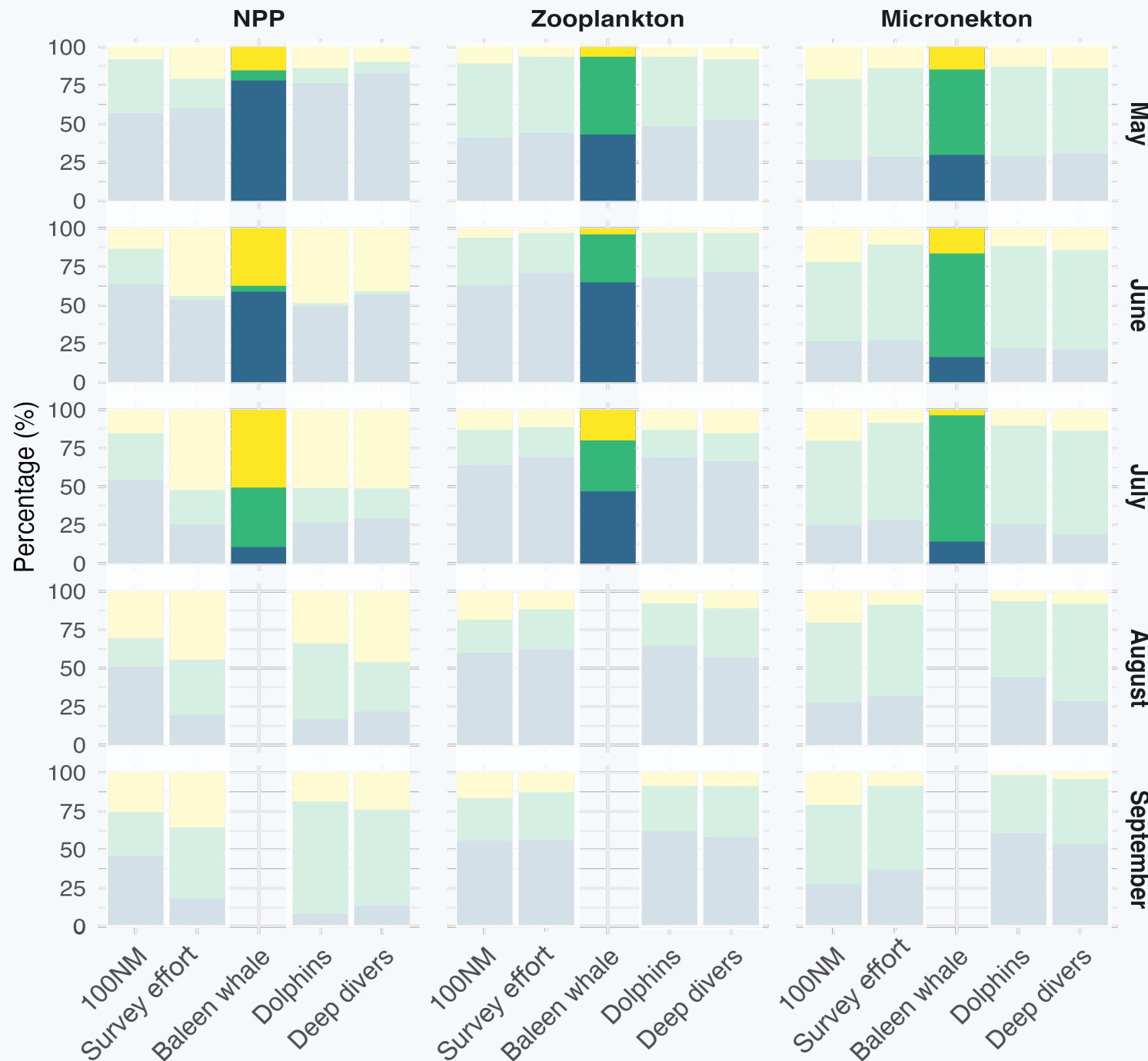
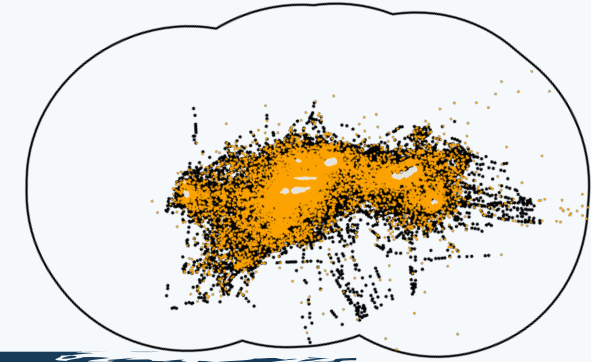


Figure 12 : Distribution of cetaceans sightings and survey effort

Results

Cetaceans and productivity



Survey effort and 100NM:

NPP:

More productive zones for the survey effort

Zooplankton and micronekton:

Same proportions

Cetaceans:

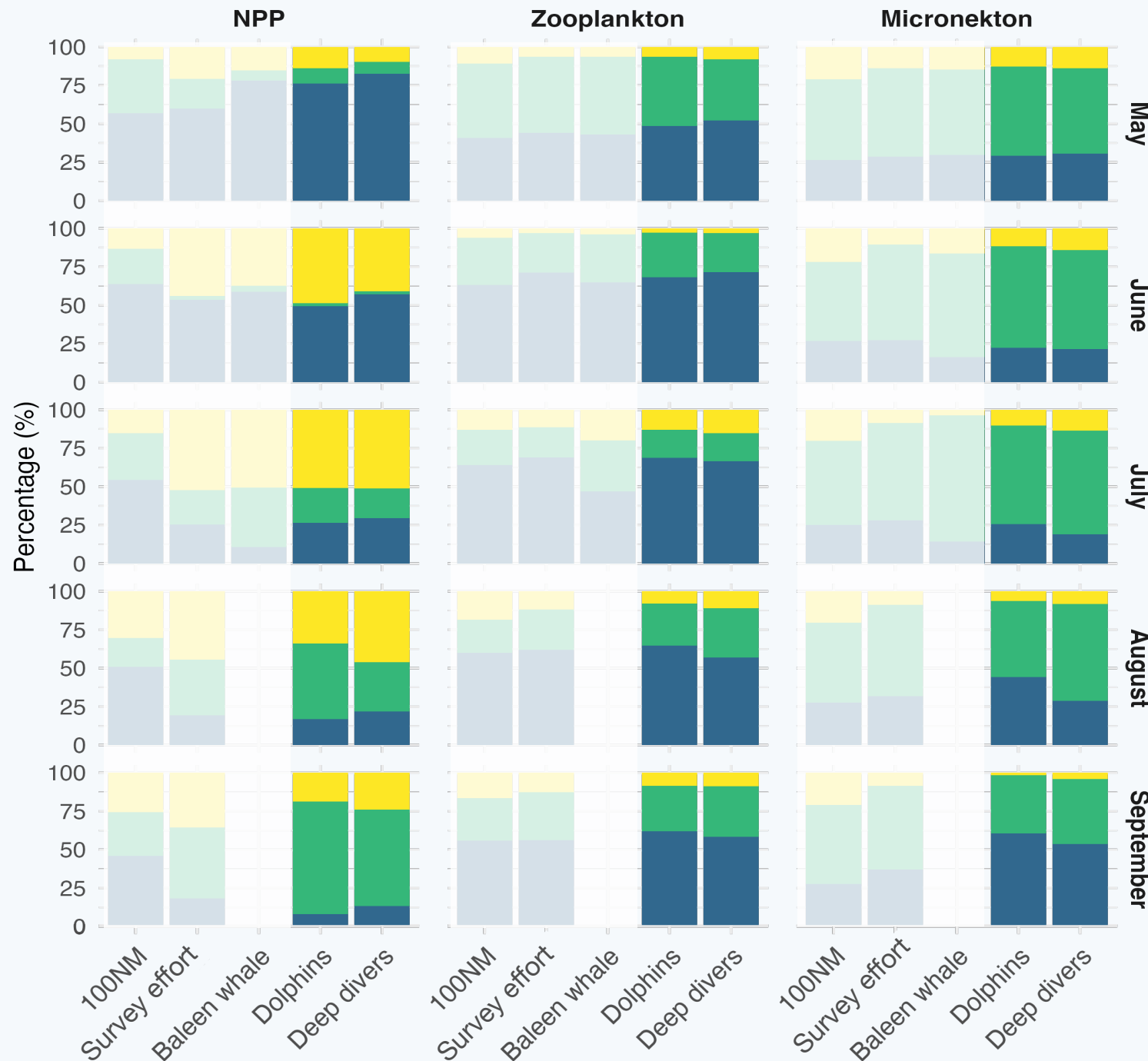
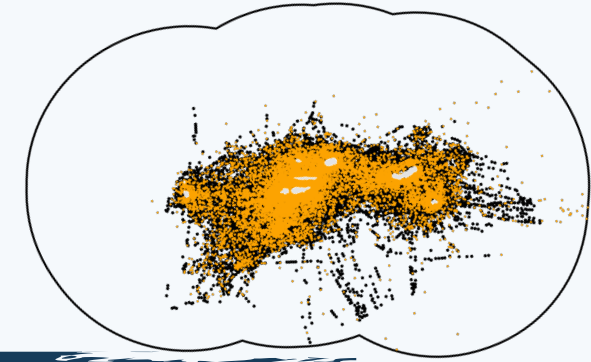
Baleen whales:

More productive zones than survey effort and 100NM

Figure 11: Proportion of each productivity zone for each variable within the 100 NM area, the vessel effort, and the three functional groups for the climatological study

Results

Cetaceans and productivity



Survey effort and 100NM:

NPP:

More productive zones for the survey effort

Zooplankton and micronekton:

Same proportions

Cetaceans:

Baleen whales:

More productive zones than survey effort and 100NM

Dolphins and deep divers:

Same proportions as the survey effort

Figure 11: Proportion of each productivity zone for each variable within the 100 NM area, the vessel effort, and the three functional groups for the climatological study

Discussion

Inter-annual variabilities



- No global trend over the 26 years
- Important inter-annuals variabilities
- Clear latitudinal gradient
 - Unproductive: south 37°N
 - Azores Counter Current

MPAs

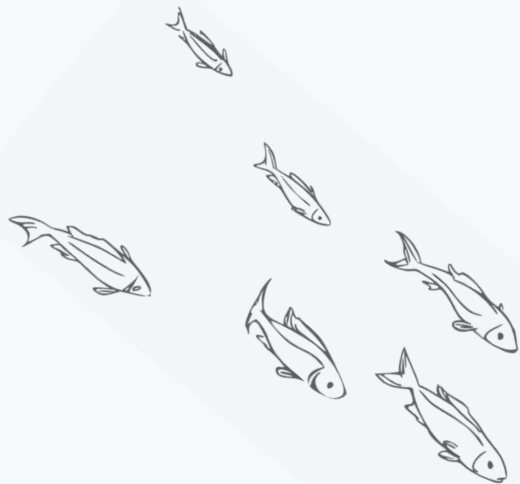


- Fully protected areas:
 - 8 months of high productivity
 - Large MPA in the north
- Highly and non protected areas
 - Similar productivity

Cetaceans



- Baleen whales:
 - Link to productive waters
 - Especially in July
- Dolphins and deep divers:
 - No clear pattern



Discussion

Limitations

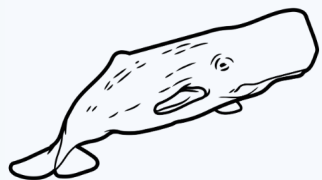
SEAPODYM-LMTL



- Developed for the Pacific Ocean → limited validation in the Central North Atlantic
- Poor performance in complex topography → micronekton aggregation and cetacean foraging

MPAs

- Designed for deep-sea habitats and seamounts → not suitable for highly mobile species
- Productivity alone is insufficient to predict cetacean presence (SST, bathymetry...)



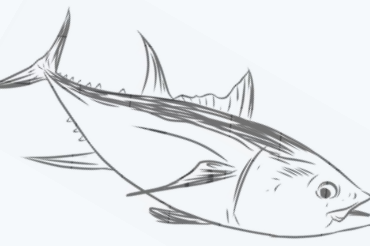
Cetaceans

- Uneven spatial coverage and limited temporal coverage (May–September)
- Species grouped into functional groups → potential dietary mismatches

Conclusion

- 
- 
- First long-term assessment of **low- and mid-trophic resources** in the Azores
 - **Latitudinal gradient**
 - > higher productivity north of **35-37°N** and around the **islands**
 - **Fully** protected areas overlaped with **productive waters** (April-November)
 - > potential habitat
 - **Baleen whales** frequently found in **productive waters**
 - **No clean pattern** for dolphins and deep divers

Perspectives

- 
- 
- **Long-term** monitoring of prey and predators
 - Combine **models** and **field observations**
 - Include additional **predictors** (fronts, seamounts,...)
 - Assess the **benefits of MPAs** for marine organisms

An underwater photograph showing four dolphins swimming in the foreground, surrounded by a massive school of small fish. The water is clear blue, and sunlight filters down from the surface. A white rectangular box with a drop shadow is centered in the image, containing the text "Thank you for your attention !".

**Thank you for
your attention !**

References

- Lehodey, P., Murtugudde, R., & Senina, I. (2010). Bridging the gap from ocean models to population dynamics of large marine predators: a model of mid-trophic functional groups. *Progress in Oceanography*, 84(1-2), 69-84.<https://doi.org/10.1016/j.pocean.2009.09.008>
- Lehodey, P., & Senina, I. (2009). A user manual for SEAPODYM (Version 2.0) application with data assimilation.
- Virgili, A., Authier, M., Monestiez, P., & Ridoux, V. (2018). How many sightings to model rare marine species distributions. *PLoS One*, 13(3), e0193231.<https://doi.org/10.1371/journal.pone.0193231>